

120 Days of Quantitative Finance

Month 3: Machine Learning for Finance

Week 10: Tree-Based Models and Market Structure

Day 68: Regime Detection with Clustering

Arnav Garg

March 2026

1 Introduction

Financial markets are not stationary. Instead, they move through different **market regimes** characterized by distinct levels of volatility, trend behavior, and risk appetite.

Examples of common regimes include:

- Bull markets with rising prices
- Bear markets with persistent declines
- High-volatility crisis periods
- Low-volatility stable environments

Identifying these regimes is important because trading strategies often perform differently under different market conditions.

One way to identify regimes is through **unsupervised learning**, specifically clustering algorithms.

2 Market Regimes

A market regime represents a statistical state of the market where the properties of returns remain relatively stable.

If r_t denotes the return at time t , then different regimes can exhibit different characteristics:

$$r_t \sim N(\mu_k, \sigma_k^2)$$

where:

- k represents the regime
- μ_k is the mean return in regime k
- σ_k^2 is the volatility in regime k

Thus, each regime corresponds to a different distribution of returns.

3 Clustering for Regime Detection

Clustering algorithms attempt to group observations based on similarity.

Given a dataset:

$$X = \{x_1, x_2, \dots, x_n\}$$

where each x_i is a feature vector describing market conditions, clustering partitions the data into K groups.

$$C_1, C_2, \dots, C_K$$

Each cluster can be interpreted as a market regime.

4 K-Means Clustering

One of the most widely used clustering algorithms is K-Means.

The objective of K-Means is to minimize the within-cluster variance:

$$\min \sum_{k=1}^K \sum_{x_i \in C_k} ||x_i - \mu_k||^2$$

where:

- C_k is the set of points in cluster k
- μ_k is the centroid of cluster k

The algorithm works iteratively:

1. Initialize K cluster centers
2. Assign each observation to the nearest center
3. Update the cluster centers
4. Repeat until convergence

5 Feature Selection for Regime Detection

To detect regimes in financial markets, we construct feature vectors that summarize market conditions.

Common features include:

- Average return
- Rolling volatility
- Trading volume
- Momentum indicators

For example, a feature vector may be defined as:

$$x_t = (r_t, \sigma_t, m_t)$$

where:

- r_t = return
- σ_t = rolling volatility
- m_t = momentum indicator

These features capture both trend and risk characteristics of the market.

6 Interpreting Clusters

After clustering the data, each cluster can be interpreted as a regime.

For example:

Cluster	Mean Return	Volatility
1	Positive	Low
2	Negative	High
3	Neutral	Medium

These clusters may correspond to:

- Bull market regime
- Crisis regime
- Sideways market regime

7 Applications in Quantitative Finance

Regime detection plays an important role in quantitative strategies.

Applications include:

- Adaptive trading strategies
- Dynamic risk management
- Portfolio allocation adjustments
- Volatility forecasting

For example, a momentum strategy may perform well in trending markets but poorly during high-volatility sideways regimes.

By detecting regimes, strategies can adapt their behavior accordingly.

8 Limitations

Although clustering provides useful insights, it has limitations:

- Choosing the correct number of clusters is challenging
- Financial regimes may evolve over time
- Clustering results may depend on feature selection

More advanced regime detection models include Hidden Markov Models and Bayesian switching models.

9 Conclusion

Market regimes are an important concept in quantitative finance. By applying clustering algorithms to financial data, we can identify periods with distinct statistical characteristics.

Understanding these regimes allows traders and researchers to build more adaptive and robust trading strategies.